

Akshansh Tyagi

akshanshtyagi2003@gmail.com | +91 9536216821

SUMMARY

Aspiring AI/ML Engineer with hands-on experience building machine learning, computer vision, and Retrieval-Augmented Generation (RAG) applications through academic and personal projects. Proficient in Python, TensorFlow, PyTorch, OpenCV, and scikit-learn, with experience in model development, data Preprocessing, feature engineering, and backend integration. Seeking AI/ML Engineer, Machine Learning Engineer, or Computer Vision Engineer opportunities.

EDUCATION

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY

2022-2026

BACHELOR OF TECHNOLOGY IN
INFORMATION TECHNOLOGY
Noida, Uttar Pradesh, India

S.D.PUBLIC SCHOOL

Muzaffarnagar, Uttar Pradesh, India
CLASS XII

Marks Obtained : 85.2

CLASS X

Marks Obtained : 89.4

SKILLS

PROGRAMMING

Experienced:

Python • C++ • SQL • MongoDB • REST
APIs • Object-Oriented Programming

Familiar:

JavaScript • Git • GitHub • React •
Node.js

AI/ML

TensorFlow • PyTorch • Scikit-learn •
OpenCV • NumPy • Pandas • YOLOv8 •
Computer Vision • Machine Learning •
Deep Learning • Retrieval-Augmented
Generation (RAG)

LINKS

GitHub: github.com/AkshanshTyagi-2003

LinkedIn: [linkedin.com/in/akshansh-tyagi-961691330](https://www.linkedin.com/in/akshansh-tyagi-961691330)

LeetCode:

leetcode.com/u/akshanshtyagi2003

EXPERIENCE

BUMIMA | SOFTWARE DEVELOPER INTERN (REMOTE)

May 2025 - Jul 2025 | Vancouver, Canada

Crypto Arbitrage Trading Engine

- Built a real-time crypto arbitrage engine in C++ using REST and WebSocket APIs to process live exchange data while reducing trade execution latency by approximately 100 ms.
- Developed automated trade execution, wallet integration, structured logging, and fault-tolerant workflows, improving backend reliability and deployment readiness.

Master - Slave Trading

- Developed a Python-based copy trading system integrating MetaTrader5 with Exness accounts, achieving approximately 95% order replication accuracy across multiple trading accounts.
- Implemented automated validation, robust error handling, structured logging, and trade synchronization to improve runtime stability and overall system reliability.

PROJECTS

YOLO-NDVI-YIELD

End-to-end crop disease detection and yield prediction pipeline using Sentinel-2 satellite imagery, YOLOv8, NDVI, and Random Forest regression.

- Processed over 2,000 Sentinel-2 image patches and extracted NDVI features for vegetation health analysis.
- Trained YOLOv8 and Random Forest models, achieving approximately 92% disease detection accuracy and an R^2 score of 0.88.
- Automated yield prediction and farming cost estimation, reducing manual analysis time by approximately 70%.

KERALA AYURVEDA INTERNAL KNOWLEDGE ASSISTANT

Corpus-grounded Retrieval-Augmented Generation (RAG) system for accurate medical knowledge retrieval with built-in safety guardrails and offline inference.

- Developed a RAG pipeline indexing over 5,000 medical knowledge chunks using semantic retrieval and document grounding.
- Implemented citation-based retrieval and safety guardrails, reducing unsupported responses by approximately 90%.
- Optimized retrieval and indexing pipelines, improving average query response time by approximately 40%.

POULTRY BIRD COUNTING SYSTEM

Real-time poultry bird detection, tracking, and counting system using YOLOv8 and ByteTrack for CCTV-based farm monitoring.

- Trained YOLOv8 on over 8,000 annotated bird images for automated detection and counting.
- Integrated ByteTrack for multi-object tracking, achieving approximately 96% counting accuracy across video streams.
- Optimized inference and visualization pipelines, improving real-time processing speed by approximately 35%.